

Benjamin Lang

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Education

Adelaide University

Jan 2025 – Present

MPhil in Applied Mathematics

- *Creating a multilayer model of Australia's Internet.*
- **Scholarships:** AMSI Summer School Travel Grant (2025), Research Training Program Stipend (2025), George Fraser Supplementary Scholarship (2025)
- **Coursework:** GPA: 7.0/7.0; WAM: 97/100

The University of Adelaide

Feb 2018 – Nov 2022

Bachelor of Mechanical and Aerospace Engineering (Honours) with Bachelor of Mathematical Sciences (Applied Mathematics)

- First-Class Honours
- GPA: 6.95/7.00; WAM: 92/100
- **Scholarships:** University of Adelaide Principal's Scholarship (2018), Joseph Miller Travel Scholarship in Engineering (2019, SEM 2)
- **Awards:** Executive Dean's Recognition of Academic Excellence (2018, 2021, 2022)

The University of Toronto

Aug 2019 – Dec 2019

Semester Abroad

- Selected to attend based on merit and written application.

Awards

Rhodes Scholarship Finalist — Rhodes Scholarship Selection Committee for SA

2023

- One of five finalists for the Rhodes Scholarship in SA. Selected based on written application, academic achievements, and demonstrations of leadership and contribution to community.

Valedictorian — School of Electrical and Mechanical Engineering

Class of 2022, SEM 2

The University of Adelaide

- Awarded based on academic merit and contribution to the School of Electrical and Mechanical Engineering.

The R.J. Jennings Memorial Prize for Mechanical Engineering Design

2022

- Awarded for receiving the top final mark for final year group honours project in mechanical engineering.

The Nova Systems Prize for Aerospace Engineering

2022

- Awarded for receiving the highest WAM within Aerospace Engineering major.

Ingenuity: 2nd Place Mechanical Engineering Project

2022

- Awarded for final year honour project, based on scoring by industry and academic judges.

Projects

University of Adelaide — Group Honours Project

2022

Regolith Resistant Lunar Rover Drivetrain

- Investigated and developed a lunar rover drivetrain more resilient to lunar regolith to support NASA's Artemis Program; developed two working prototype subsystems: a magnetic transmission system and electrodynamic dust shield (EDS). Submitted and presented at ICIUS 2023.

Adelaide Summer Research Scholarship (ASRS)

Dec 2021 – Feb 2022

University of Adelaide — School of Mathematical Sciences

- *(Mis)information theory, influence and emotional contagion on social media.*

- Utilised NLP techniques in Python to investigate Australian media subtitles and develop topic models.

Experience

Graduate Data Scientist

Aurizon

Adelaide, SA

Mar 2024 – Present

- Developing and integrating current methods for multi-modal fusion of different data sources, utilising open-source tools to create a data lakehouse. Creating applications for data extraction from text using LLMs.

Tutor/Marker

Adelaide University

Adelaide, SA

Mar 2021 – Present

- Systems Engineering and Industry Practice: tutoring and report marking (*SEM 1 2021 – SEM 2 2025*)
- Data Skills for Scientists: report marking (*SEM 1 2026*)
- Data Science Research Project Part A: report marking (*SEM 1 2026*)
- Engineering Modelling 1: tutoring and assignment marking (*SEM 2 2026*)
- Engineering Modelling 2: assignment marking (*SEM 2 2026*)

Graduate Systems Engineer

BAE Systems

Edinburgh, SA

Feb 2023 – Feb 2023

- Assisted with major design reviews and upgrades of Defence projects; completion of configuration management database migrations; developed interactive budgets for SWAP-C management across worksites.

STEM Cadet

Defence Science and Technology Group (DSTG)

Edinburgh, SA

Feb 2022 – Feb 2023

- Undertook a STEM cadetship program as part of the Passive Radar team; developed signal processing algorithms and developed novel methods for target localisation using bi-static radar.

Volunteering

Secretary/Committee Member

Old Concordians Soccer Club

Sep 2018 – Present

- Involved in founding the Old Concordians Soccer Club in 2019. Have played for and helped run the club ever since. Current acting secretary of the club; facilitating meetings and organising club correspondence.
- Organising club events; managing training and match-day logistics, including equipment, kits, which must be split between four teams; managing match-day and training coordinators.

Peer Assisted Study Sessions (PASS) Leader

The University of Adelaide

Mar 2019 – Jul 2022

- Selected as PASS Leader for Mathematics IA, Mathematics IB, and Engineering Mechanics – Dynamics.
- Involved hosting weekly sessions containing up to 20 students. Facilitated sessions encouraging students to learn together with fun group activities.
- Witnessed significant improvement in student's study skills and understanding of complex topics.

Publications

Vine Hall, H., Mortier-Spole, O., Nguyen, V., **Lang, B.**, Laden, M., Munn, M., Robertson, W., “Investigating the Use of Electrodynamical Dust Shielding to Improve the Robustness of a Lunar Rover Drivetrain to Regolith”, in *Proceedings of the 19th International Conference on Intelligent Unmanned Systems*, The University of Adelaide, Adelaide, Jul. 2024, pp. 179-189.

[10.1007/978-981-97-6591-1_17](https://doi.org/10.1007/978-981-97-6591-1_17) [🔗](#)

Lang, B., Roughan, M., and Ye, M., “Robustness Analysis of Australia’s Internet Using a Multilayer Network Model”, arXiv preprint, 2026.

[arXiv:2510.13330](https://arxiv.org/abs/2510.13330) [🔗](#)

Conference Presentations

MODSIM 2025

Oral Presentation: Network analysis of Australian Internet traffic dynamics.

Adelaide, Australia

Dec 2025

NetSciX 2026

Poster: Reliability assessment of Australia's Internet infrastructure using a multi-layer network model.

Auckland, New Zealand

Feb 2026